gpeyt1-31 · Corrosion allowance

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

Uncoated carbon-steel H-piles will be driven through a fill with field resistivity 1,500 Ω·cm (pH 5.8, sulfates 400 ppm) for a bridge with a 75-year design life. The owner requires the maximum (conservative) loss rate per face. Flange thickness is 0.61 in.

Most nearly, the sacrificial thickness loss per exposed face over the design life is:

A. 0.099 in.
B. 0.049 in.
C. 0.016 in.
D. 0.187 in.

Given equations / data

> $x = K\,t^{n}$ (μm); carbon steel $n = 0.80$, $K = 40$ (avg) / 80 (max); 1,000 μm = 1 mm = 0.039 in.

Worked solution

1. 1,500 Ω·cm → corrosive (700–2,000) per resistivity table
2. x = 80 × 75^0.80 = 2,530 μm
3. = 2.53 mm × 0.039 = 0.099 in. per face

Final answer

0.099 in.

Why the other values are wrong

- **0.049 in.** — used the average K = 40
- **0.016 in.** — used galvanized-steel constants
- **0.187 in.** — multiplied t by n instead of raising to n

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.17.1 Reactive and Corrosive Soils (resistivity table), p. 197; §3.17.2 Corrosion of Buried Steel, p. 198

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZOpOWZXWblU (no creator values copied).

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